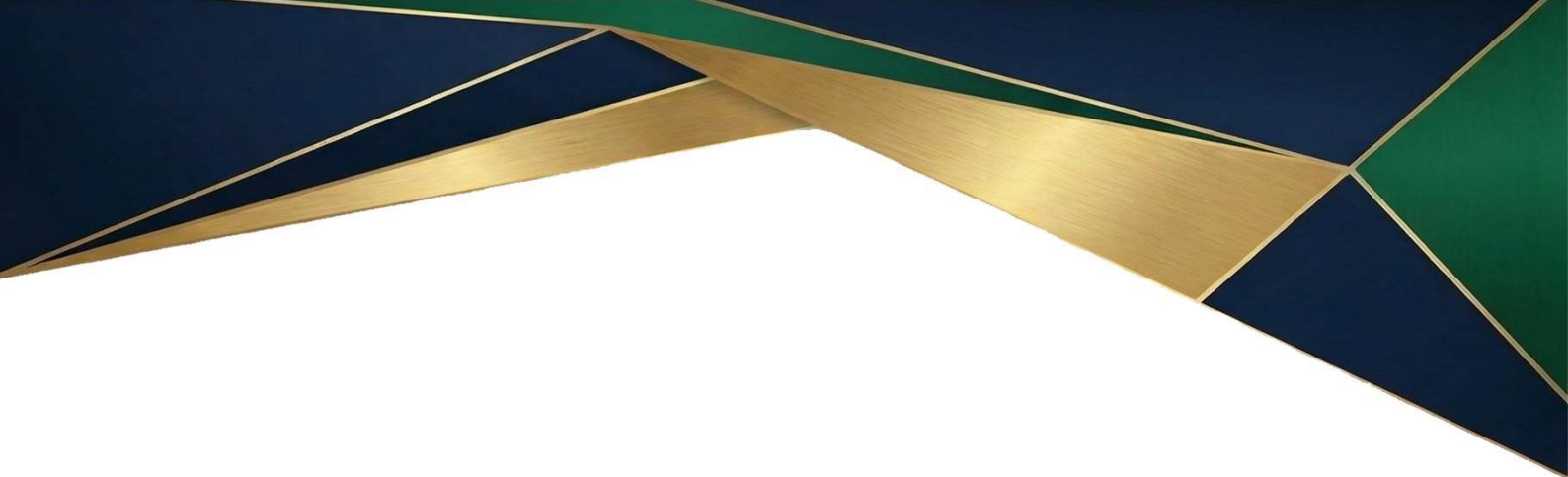




Optron JSC

Product Catalog

2026

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About the Company



JSC "Optron" traces its history to 1928, when the Krasin Radio Plant was established by decision of the Supreme Soviet of the RSFSR to manufacture radio receivers.

The initial organizational stage of production at the site now occupied by JSC "Optron" dates back to 1928-1938. At that time, three enterprises were established on the present territory from Fortunatovskaya Street to Mironovskaya Street: the Krasin Radio Plant (1928), successor to a factory producing nickel-plated beds; the Koopputilovets artel (1933), focused on consumer goods; and the Sirocco plant, established in 1938 to produce centrifugal fans.

In 1938, the radio plant was renamed the Krasin Stamping and Mechanical Plant for stainless-steel spoons, and retained this name until 1949.

During the Great Patriotic War, the enterprises produced military products: grenades and aircraft units, refuelling vehicles based on the ZIS-5 chassis, and gearboxes for Katyusha rocket launchers; they belonged to local industry.

In 1945, the Koopputilovets artel and the Sirocco plant were merged into Plant No. 20 and transferred to the Ministry of Communications Equipment Industry.

The plant was assigned the production of ceramic parts from alundum powder, while the Krasin plant developed and launched production of Rekord loudspeakers and the Salyut radio console, which in 1946 received second place from the Chamber of Commerce among the best radio-engineering products.

In 1953, the enterprise mastered the production of control gear for fluorescent lighting in the metro and residential buildings. The plant also produced alundum lamp heaters.



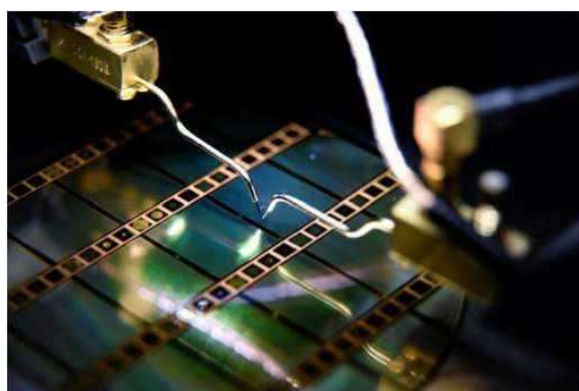
About the Company



On the basis of Resolution No. 3482-1620 of the USSR Council of Ministers dated 14 Sept 1951 and Order No. 437 of the CPSU Central Committee, Research Institute 311, with Pilot Plant 311 and a design bureau for the development and introduction of vacuum electronic devices, was organized on the basis of Plant 311. In 1956, a new area of activity was defined for Plant No. 311 - the production of semiconductor devices.



In subsequent years, Plant No. 311 was reorganized and renamed as the plant under the Research Institute of Semiconductor Device Engineering, the plant under the Saphir Research Institute, the State Plant "Optron", OJSC "Optron" (1994), and later OJSC "Optron".



The company's core activity is the development and production of electronic components for the radio-electronic industry, including Zener diodes, RF variable and switching diodes, microwave switching and limiting diodes, matrix indicators, hybrid integrated circuits, and related products.



About the Company

The enterprise has production facilities and a young, highly qualified workforce of workers and engineering specialists.



The achievements of the JSC "Optron" team have been recognized with the Lenin Prize and USSR State Prizes, and many employees have been awarded orders and medals.

JSC "Optron" specializes in the development and production of epitaxial structures in the spectral range from ultraviolet to near infrared, as well as in the development and manufacture of widely used electronic components based on these materials.



Licenses and Certificates



Certificate of Conformity
No. ES 06.093.0474-2025



Patent No. 174672
Title: "Device for Measuring Static Parameters of Semiconductor Devices"



Patent No. 2572503
Title: "Method for Depositing a Single-Crystal Cubic Boron Nitride Film on a Semiconductor Silicon Substrate"



Patent No. 2377699
Title: "Light-Emitting Element"

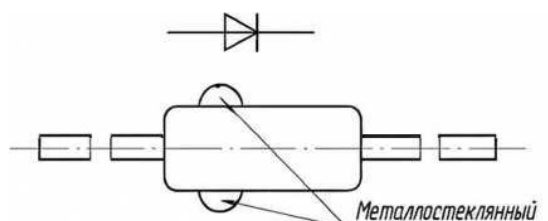
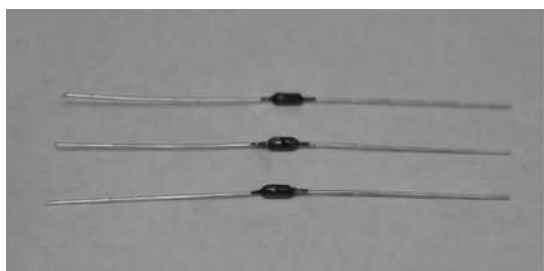


Diodes



Variable resistive diode 2D413A, 2D413B

Silicon epitaxial with a p-i-n structure variable resistive RF diodes types 2D413A and 2D413B in a metal-glass package, intended for operation as controlled RF resistive elements in special-purpose equipment.



Diode type	Number of marking dots	Color of marking dots
2D413A	One	green
2D413B	Two	green and red

MAIN ELECTRICAL PARAMETERS at temperature ambient temperature = $25 \pm 10^\circ\text{C}$

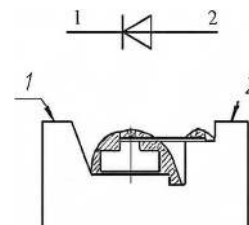
Parameter name, unit of measurement, measurement mode	Symbol designation	2D413A		2D413B	
		min.	max.	min.	max.
DC forward voltage ($I_{\text{fwd}}=20\text{mA}$), V	U_{fwd}	—	1	—	1
Total diode capacitance ($U_{\text{rev}}=0\text{V}$), pF	C_d	—	0,7	—	0,7
Differential resistance ($I_{\text{fwd}}=2\text{mA}$, $f=50\text{MHz}$), Ohm	r_{dif}	30	60	40	80
Recovery charge ($I_{\text{fwd}}=2\text{mA}$, $U_{\text{rev}}=10\text{V}$), nC	Q_{rec}	2	—	2	—
Breakdown voltage ($I_{\text{rev}}=500\text{uA}$) V	U_{br}	30	—	30	—

Weight not more than 0,035 g



Switching diode 2D531A-6, 2D531B-6

Chip-type silicon mesa-epitaxial with a p-i-n structure switching semiconductor diodes, intended for use as part of hybrid integrated circuits 30-400 MHz frequency range



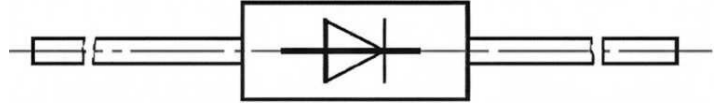
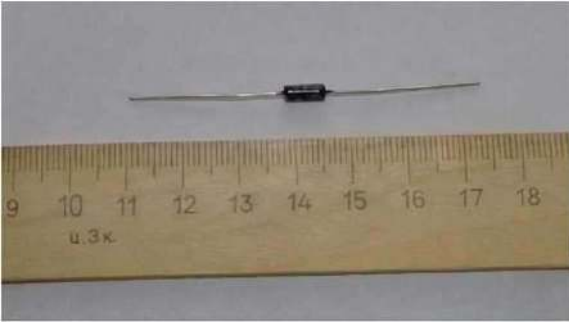
Main electrical parameters

Parameter name, (measurement mode), unit of measurement	Symbol designation	Limit				Temperature , °C
		2D531A-6		2D531B-6		
		Min.	Max.	Min.	Max.	
Total diode capacitance, pF Urev=50 V, fmeas=10 MHz	Cd		0,6		0,6	25±10
Forward loss resistance, Ohm Ifwd=20 mA, fmeas=50 MHz	r_fwd		0,6		0,6	25±10
Breakdown voltage, V Ifwd=10 uA	Ubr	150		150		25±10
DC forward voltage, V Ifwd=20 mA	Ufwd		1		1	25±10
Reverse loss resistance, kOhm Urev=40 V, fmeas=50MHz	Rrev	600				25±10
Stored charge, Ifwd=20 mA, Urev=100 V	Qst	10				25±10
2nd and 3rd harmonic coefficient under forward bias, dB Ifwd=65 mA, fmeas=50 MHz Pmeas=200 mW						
2nd harmonic	K2,fwd		-100			
3rd harmonic	K3,fwd		-100			
2nd and 3rd harmonic coefficient under reverse bias, dB Urev=40 V, fmeas=50 MHz, fmeas=300 MHz, Pmeas=200mW						
2nd harmonic	K2,rev		-90			
3rd harmonic	K3,rev		-90			

Weight not more than 0,03 g



Switching diode 2D420A



P-i-n diodes type 2D420A silicon epitaxial-planar switching in an all-glass package, intended for switching high-frequency signals in frequency range from 30 to 300 MHz in special-purpose equipment.

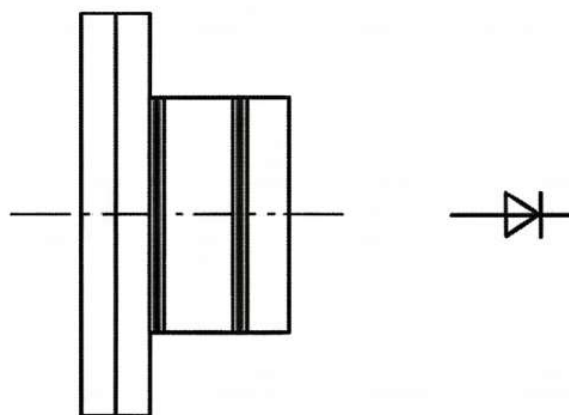
Main electrical parameters

Parameter name, (measurement mode), unit of measurement	Symbol designation	Limit	
		Min.	Max.
DC forward voltage at $I_{fwd}=50$ mA at temperature ambient temperature 25 ± 10 °C, V	U_{fwd}	—	1,0
Differential resistance at $I_{fwd}=10$ mA and $f=50$ MHz, at temperature ambient temperature 25 ± 10 °C, Ohm	r_{dif}	—	1,0
Breakdown voltage ($I_{rev}=10$ uA) at temperature ambient temperature 25 ± 10 °C, V	U_{br}	35	—
Total diode capacitance at $U_{rev}=0$ V, $f=10$ MHz, pF	C_d	—	1,0

Weight not more than 0,25 g



Limiting diode 2A534A, 2A534B, KA534A, KA534B



Semiconductor microwave diodes 2A534A,B, and KA534A,B silicon planar-epitaxial limiting in a metal-ceramic package KD-102, intended for use in devices receiver protection, stabilization, limiting and adjustment of microwave power level in centimeter and decimeter wavelength ranges.

Main electrical parameters

Parameter name (measurement mode), unit of measurement	Symbol designation	Limit				Ambient temperature, °C
		2A534A (KA534A)		2A534B (KA534B)		
		Min.	Max.	Min.	Max.	
Differential resistance at Ifwd=100mA, Ohm	r_dif	—	1,8	—	2,5	25±10
Diode resistance at low microwave power level P<1 mW, Ohm	r_low	—	10	—	15	25±10
			10		15	135±5
			15		20	60±3
Rated reverse voltage at Irev=10uA, V	Unorm, rev.	30	110	30	110	25±10
Stored charge in switching mode with forward current 10mA at pulse reverse voltage 10V, nC	QHK	—	1	—	1	25±10
Diode package inductance, nH	L	—	0,3	—	0,3	25±10
Package capacitance diode, pF	Cpkg	0,18	0,26	0,18	0,26	25±10
Total diode capacitance at Urev=0 V, pF	Cd	0,4	0,65	0,35	0,5	25±10

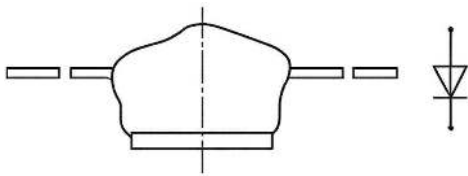
Weight not more than 0,05 g



Limiting chip diode 2A522A-2



Silicon planar-epitaxial limiting diode microwave 2A522A-2, intended for use in special-purpose equipment centimeter and decimeter wavelength range as part of hybrid integrated circuits, units and equipment, providing sealing and protection of devices from moisture exposure, salt fog, mold fungi, frost and dew, aggressive gases and mixtures.



Main electrical parameters

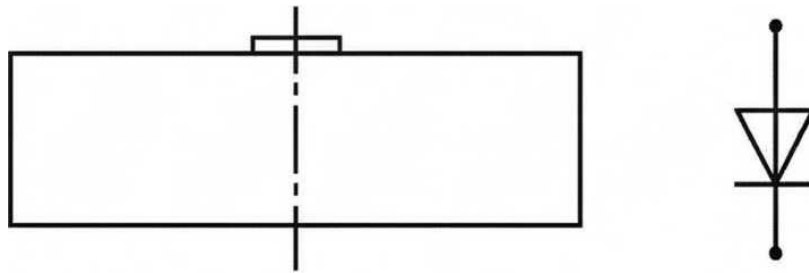
Parameter name (measurement mode) unit of measurement	symbol designation	Limit	
		Min.	Max.
Differential resistance diode at $I_{fwd}=100\text{mA}$, Ohm	r_{dif}	—	1,8
Total capacitance at $U_{rev}=0\text{V}$, pF	C_d	0,35	0,75
Diode resistance at low microwave power level at temperature ambient temperature= $25\pm 10^\circ\text{C}$, Ohm at temperature ambient temperature= $125\pm 5^\circ\text{C}$, Ohm at temperature ambient temperature= $-60\pm 3^\circ\text{C}$, Ohm	r_{low} r_{low} r_{low}	— — —	12 17 12
Rated DC reverse voltage at $I_{rev}=100\mu\text{A}$, V	$U_{norm,rev}$	5	70
Stored charge diode at $I_{fwd}=50\text{mA}$, nC	Q_{HK}	—	10

Weight not more than 0,004 g

Note - Differential resistance (r_{dif}) is numerically equal to the diode resistance at high microwave power level.



Limiting chip diode 2A522A-5



Silicon planar-epitaxial limiting diode microwave 2A522A-5, intended for use in special-purpose equipment centimeter and decimeter wavelength range as part of hybrid integrated circuits, units and equipment, providing sealing and protection of devices from moisture exposure, salt fog, mold fungi, frost and dew, aggressive gases and mixtures.

Main electrical parameters

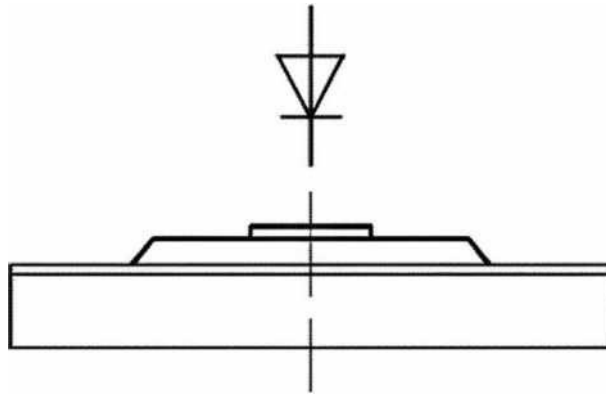
Parameter name (measurement mode) unit of measurement	symbol designation	Limit	
		Min.	Max.
Differential resistance diode at $I_{fwd}=100\text{mA}$, Ohm	r_{dif}	—	1,8
Total capacitance at $U_{rev}=0\text{V}$, pF	C_d	0,2	0,6
Diode resistance at low microwave power level, Ohm at temperature ambient temperature $=25\pm 10^\circ\text{C}$, Ohm at temperature ambient temperature $=125\pm 5^\circ\text{C}$, Ohm at temperature ambient temperature $= -60\pm 3^\circ\text{C}$, Ohm	r_{low} r_{low} r_{low}	— — —	12 17 12
Rated DC reverse voltage at $I_{rev}=100\mu\text{A}$, V	$U_{norm,rev}$	5	70
Stored charge diode at $I_{fwd}=50\text{mA}$, nC	Q_{HK}	—	10

Weight not more than 0,001 g

Note - Differential resistance (r_{dif}) is numerically equal to the diode resistance at high microwave power level.



Limiting chip diode 2A550A-5



Microwave diodes type 2A550A-5 chip-type silicon planar-epitaxial r+-n -n+ limiting, intended for internal mounting in hybrid integrated circuits, microassemblies and units in equipment microwave

Main electrical parameters

Parameter name, (measurement mode), unit of measurement	Symbol designation	Limit		Temperature, °C	Note
		Min.	Max.		
Breakdown voltage at reverse current 10 uA, V	U _{br}	100		25±10	
Stored charge at I _{fwd} =10 mA, U _{rev} =10V, nC	Q _{HK}	0,3 0,4 0,2	1 1,5 1	25±10 135±5 -60±3	1
DC forward voltage at I _{fwd} -100 mA, V	U _{np}	0,85	1,1	25±10	
Resistance at low microwave power level (at zero bias) P _{low} =1 mW, F=4.3 GHz, Ohm	r _{low}		30	25±10	1
Structure capacitance at zero bias f = 10 MHz, pF	C _{str 0}	0,2	0,6	25±10	
Structure capacitance at reverse bias U _{rev} =80 V. f = 10 MHz, pF	C _{str}	0,1	0,5	25±10	
Leakage power in the range input power up to 2.5 W, W	P _{leak}	-	0,8	25±10	1,2

Weight not more than 0,001 g

Notes

1. Measurements and tests are performed in a reference metal-ceramic package KD-105.
2. Measurements are performed at a frequency of $3 \pm 10\%$ GHz in a resonant chamber with a 50-ohm line.



Microwave diode 2A560A

Diodes semiconductor microwave limiting silicon planar-epitaxial p+-n-n+ type 2A560A in metal-ceramic package, intended for operation in protected devices frequency range 3-11 GHz in special-purpose equipment.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Breakdown voltage at reverse current 10 uA, V	Ubr	55	150	25±10
Effective lifetime of nonequilibrium charge carriers, ns, Ifwd=10mA; Urev=10 V	tau_eff		80	25±10
Rated DC reverse voltage, V, Irev≤5 uA	Unorm	50		25±10
DC forward voltage, V, Ifwd=100mA	Ufwd	0,85 0,9 0,75	1,2 1,25 1,1	25±10 -60±3 125±3
Forward loss resistance, Ohm, Ifwd=10mA; f=500 MHz	r_fwd		1,5	25±10
Total diode capacitance, pF, Urev=30V	Cd	0,3	0,45	25±10
Thermal resistance, junction-to-case, °C/W (single-sided heat removal)	Rth j-c		30	25±10
Package capacitance, pF	Cpkg	0,21	0,25	25±10
Diode inductance, nH	Ln		0,5	25±10
Thermal relaxation time, us	tau_t	20		25±10

Weight not more than 0,2 g



Microwave diode 2A560A-5

Diodes semiconductor microwave limiting silicon planar-epitaxial p+-n-n+ type 2A560A-5 chip-type, intended for operation in protected devices frequency range 3-11 GHz in special-purpose equipment.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Breakdown voltage at reverse current 10 uA, V	Ubr	55	150	25±10
Effective lifetime of nonequilibrium charge carriers, ns, Ifwd=10mA; Urev=10 V	tau_eff		80	25±10
Rated DC reverse voltage, V, Irev≤5 uA	Unorm	50		25±10
DC forward voltage, V, Ifwd=100mA	Ufwd	0,85 0,9 0,75	1,2 1,25 1,1	25±10 -60±3 125±3
Forward loss resistance, Ohm, Ifwd=10mA; f=500 MHz	r_fwd		1,5	25±10
Structure capacitance, pF, Urev=30V	Cstr	0,05	0,2	25±10
Thermal resistance, junction-to-case, °C/W (single-sided heat removal)	Rth j-c		30	25±10
Diode inductance, nH	Ln		0,5	25±10
Thermal relaxation time, us	tau_t	20		25±10

Weight not more than 0,001 g



Switching diode 2A543A-5, 2A543A-6

Microwave diodes chip-type, silicon mesa-epitaxial, switching type 2A543A-5, 2A543A-6, intended for use as part of hybrid integrated circuits 0,3-18 GHz frequency range.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Total diode capacitance, pF, $U_{rev}=20V$; $f=10$ MHz	Cd	0,12	0,19	25±10
Critical frequency, GHz, $I_{fwd}=5mA$; $U_{rev}=20V$	Fcrit	300	-	25±10
Stored charge, nC, $I_{fwd}=5mA$	Qst	0,5	3	25±10
DC forward voltage, V, $I_{fwd}=5mA$	Ufwd	-	0,9 0,9 1,2	25±10 125±3 -60±3
Breakdown voltage at reverse current 10 uA, V	Ubr	100	-	25±10
Forward loss resistance, Ohm, $I_{fwd}=5mA$	r_fwd	-	1,5	25±10

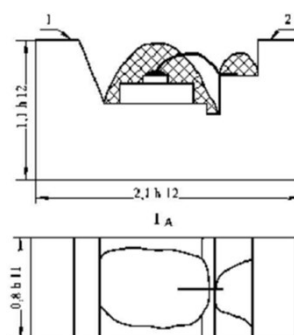
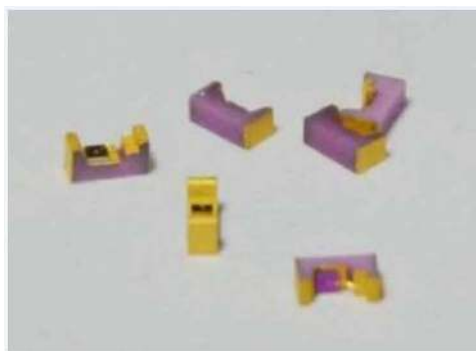
Weight not more than 0,03g-A-6;
0,015g-A-5



Switching diode 2A543B-6

Chip-type silicon mesa-epitaxial with a p-i-n structure switching semiconductor
Microwave diodes type 2A543B-6, intended for use as part of hybrid integrated circuits
0,3-18 GHz frequency range.

Diodes type 2A543B-6 are manufactured in UHL climatic version for use in non-hermetically sealed hybrid circuits.



Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Total diode capacitance, pF, $U_{rev}=20V$; $f=10\text{ MHz}$	Cd	0,15	0,22	25 ± 10
Critical frequency, GHz, $I_{fwd}=5mA$; $U_{rev}=20V$	Fcrit	300	-	25 ± 10
Stored charge, nC, $I_{fwd}=5mA$	Qst	0,5	3	25 ± 10
DC forward voltage, V, $I_{fwd}=5mA$	Ufwd	-	0,9 0,9 1,2	25 ± 10 125 ± 3 -60 ± 3
Breakdown voltage at reverse current 10 μA , V	Ubr	100	-	25 ± 10
Forward resistance, Ohm, $I_{fwd}=5mA$	R_fwd	-	1,5	25 ± 10

Weight not more than 0,01 g



Switching diode 2A542A1, 2A542B2, 2A542V2

2A542A1, 2A542B2, 2A542V2 switching silicon epitaxial semiconductor Microwave diodes in a metal-ceramic package KD-105, intended for operation in switching devices microwave wave range in special-purpose equipment.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2A542A1		2A542B2		2A542V2		Temperature , °C
		Min.	Max.	Min.	Max.	Min.	Max.	
Total diode capacitance, pF, Urev=100V;	Cd	0,55	1	0,55	1	0,55	1	25±10
Critical frequency, GHz, Ifwd=100mA; Urev=100V	Fcrit	250		250		250		25±10
Stored charge, nC, Ifwd=100mA	Qst		400 650 400		400 650 400		400 650 400	25±10 125±3 -60±3
DC forward voltage, V, Ifwd=100mA	Ufwd		1,1		1,1		1,1	25±10
Breakdown voltage at reverse current 100 uA, V	Ubr	1100		1150		1400		25±10
Forward loss resistance, Ohm, Ifwd=100mA	R_fwd		1,7		1,6		1,6	25±10

Weight not more than 0,5 g



Switching diode 2A541A-6, 2A541B-6

Silicon epitaxial switching semiconductor Microwave diodes types 2A541A-6, B-6 intended for use as part of hybrid integrated circuits, units and equipment microwave special-purpose, providing tightness and protection of devices from moisture exposure, salt fog, mold fungi, aggressive gases and mixtures.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2A541A-6		2A541B-6		Temperature, °C
		Min.	Max.	Min.	Max.	
Total diode capacitance, pF	Cd	0,15	0,22	0,18	0,25	25±10
Critical frequency, GHz, Ifwd=100mA; Urev=100V	Fcrit	400		400		25±10
Stored charge, nC, Ifwd=100mA	Qst	60	150	60	150	25±10
Breakdown voltage at reverse current 10 uA, V	Ubr	300		300		25±10
Forward loss resistance, Ohm, Ifwd=100mA	R_fwd		1,3		1,3	25±10

Weight not more than 0,2 g

* - manufactured by the enterprise on customer order and under contract



Switching diode 2A537A, KA537A, KA537B



Semiconductor microwave diodes switching, silicon diffused type 2A537A, KA537A, KA537B in a metal-ceramic package type KT-16-1 (GOST R 57439) with insulated heat sink intended for operation in switching devices santimetrovogo, decimeter, meter, and HF wave ranges.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2A537A, KA537A		KA537B		Temperature, °C
		Min.	Max.	Min.	Max.	
Total diode capacitance, pF	Cd		3		3	25±10
Critical frequency, GHz, Ifwd=100mA; Urev=100V	Fcrit	200		100		25±10
Stored charge, nC, Ifwd=100mA	Qst	400	1000	200	1500	25±10
		400	1500	200	1500	125±3
		200	1000	100	2000	-60±3
Breakdown voltage at reverse current 10 uA, V	Ubr	600		300		25±10
Forward loss resistance, Ohm, Ifwd=100mA	R_fwd		0,5		1	25±10

Weight not more than 3 g



Switching diode 2A520A, KA520A, KA520B

Microwave diodes switching, silicon epitaxial with a p-i-n structure in a metal-ceramic package KD-105, intended for operation in switching devices centimeter and decimeter wave ranges.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2A520A, KA520A		KA520B	
		Min.	Max.	Min.	Max.
Total diode capacitance, pF Urev=100V	Cd	0,4	1	0,4	1
Critical frequency, GHz, Ifwd=100mA; Urev=100V At temperature: 25±10 °C 125±5 °C -60±3 °C	Fcrit	200 170 170		150 120 120	
Stored charge, nC, Ifwd=100mA; Urev=300V	Qst		300		300
Breakdown voltage at reverse current 100 uA, V	Ubr	800		600	
Forward resistance, Ohm, Ifwd=100mA; At temperature: 25±10 °C 125±5 °C -60±3 °C	r_fwd		2 2,3 2,3		3 3,3 3,3
Package capacitance, pF	Cpkg	0,3	0,45	0,3	0,45

Marking

Diode type	Marking dot
2A520A	black, red
KA520A	black, red, green
KA520B	yellow, red, green

Weight not more than 1,3 g



Switching diode 2A515A, KA515A

Microwave diodes switching, silicon epitaxial with a p-i-n structure in a metal-ceramic package type KD-105, intended for operation in switching devices short-wave part of the centimeter wavelength range.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Min.	Max.
Total diode capacitance, pF $U_{rev}=50V$	Cd	0,4	0,7
Critical frequency, GHz, $I_{fwd}=100mA$; $U_{rev}=50V$ At temperature: 25±10 °C 125±5 °C -60±3 °C	Fcrit	100 70 70	
Stored charge, nC, $I_{fwd}=25mA$; $U_{rev}=100V$	Qst		15
Breakdown voltage at reverse current 100 uA, V	Ubr	100	
Forward resistance, Ohm, $I_{fwd}=25mA$;	R_fwd		2,5
Package capacitance, pF	Cpkg	0,3	0,45

* - manufactured by the enterprise on customer order and under contract

Marking

Diode type	Marking dot
2A515A	green
KA515A	green, green

Weight not more than 1,3 g



Switching diode 2A509A, 2A509B, KA509A, KA509B

Microwave diodes switching, silicon epitaxial with a p-i-n structure in a metal-ceramic package type KD-105, intended for operation in switching devices centimeter and decimeter wavelength ranges.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2A509A, KA509A		2A509B, KA509B	
		Min.	Max.	Min.	Max.
Total diode capacitance, pF Urev=100V	Cd	0,9	1,2	0,7	1,0
Critical frequency, GHz, Ifwd=100mA; Urev=100V At temperature: 25±10 °C 125±5 °C -60±3 °C	Fcrit	150 100 100		150 100 100	
Stored charge, nC, Ifwd=100mA; Urev=100V	Qst		25		25
Breakdown voltage at reverse current 10 uA, V	Ubr	200		200	
Forward loss resistance, Ohm, Ifwd=100mA;	R_fwd		1,5		1,5
Package capacitance, pF	Cpkg	0,3	0,45	0,3	0,45

Marking

Diode type	Color and symbol
2A509A	red dot
2A509B	blue dot
KA509A	red dash
KA509B	blue dash

Weight not more than 1,3 g



Switching diode 2A507A, 2A507B KA507A, KA507B

Microwave diodes switching, silicon epitaxial with a p-i-n structure in a metal-ceramic package type KD-105, intended for operation in switching devices centimeter and decimeter wavelength ranges.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2A507A, KA507A		2A507B, KA507B	
		Min.	Max.	Min.	Max.
Total diode capacitance, pF Urev=100V	Cd	0,8	1,2	0,8	1,2
Critical frequency, GHz, Ifwd=100mA; Urev=100V At temperature: 25±10 °C 125±5 °C -60±3 °C	Fcrit	200 180 180		200 180 180	
Stored charge, nC, Ifwd=100mA; Urev=100V	Qst		200		200
Breakdown voltage at reverse current 10 uA, V	Ubr	500		300	
Forward loss resistance, Ohm, Ifwd=100mA;	R_fwd		1,5		1,5
Package capacitance, pF	Cpkg	0,3	0,45	0,3	0,45

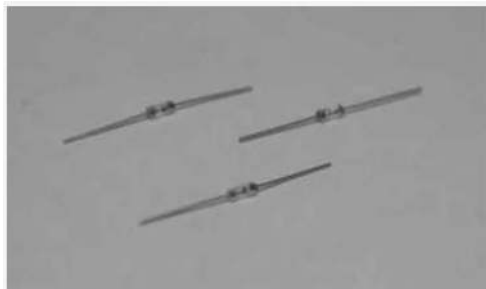
Marking

Diode type	Marking dot
2A507A	black
2A507B	yellow
KA507A	two black
KA507B	two yellow

Weight not more than 1,3 g



Mixer diodes 2A120A, 2A120AG, 2A120AR



Microwave diode mixer, silicon planar-epitaxial with Schottky barrier in a package type KD-1, intended for operation in the range 0,3-18 GHz.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit	
		Min.	Max.
Conversion loss 2A120A, dB at temperature: 25±10 °C 125±5 °C -60±3 °C	Lconv		5,5 6,5 6,0
Normalized noise figure, dB	Fnorm		7
Output resistance, Ohm	rout	200	500
Voltage standing wave ratio	VSWR		1,8
Rectified current, mA	Irect	0,5	2,5

Weight not more than 0,035 g

2A120AG(quad sets) and 2A120AR(pair) spread $U_{fwd} \leq 20$ mV at $I_{fwd} = 1$ mA and darkperature 25±10 °C



Diodes 2D419A, 2D419B, 2D419V

Silicon n-type conductivity epitaxial-planar universal diodes with Schottky barrier in a glass package, intended for detection of IF-range signals in linear detector circuit in special-purpose equipment.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	2D419A		2D419B		2D419V		Temperature, °C
		Min.	Max.	Min.	Max.	Min.	Max.	
DC reverse voltage at current 10 μ A, V	Urev	15 7,5 10		30 15 24		50 25 40		25 $\pm$ 10 -60 $\pm$ 3 125 $\pm$ 5
DC forward voltage at current 1 mA, V	Ufwd		0,4 0,5 0,4		0,4 0,5 0,4		0,4 0,5 0,4	25 $\pm$ 10 -60 $\pm$ 3 125 $\pm$ 5
Total diode capacitance, pF Urev=0V	Cd		1,5		1,5		1,5	
DC forward voltage at current 0,1 mA, V	Ufwd	0,15		0,15		0,15		

Weight not more than 0,035 g

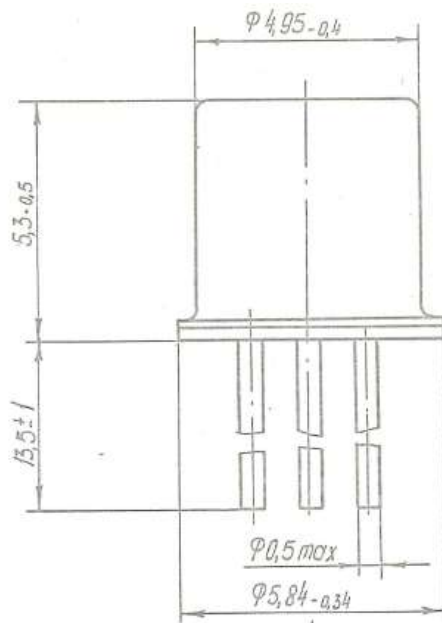


Microcircuits

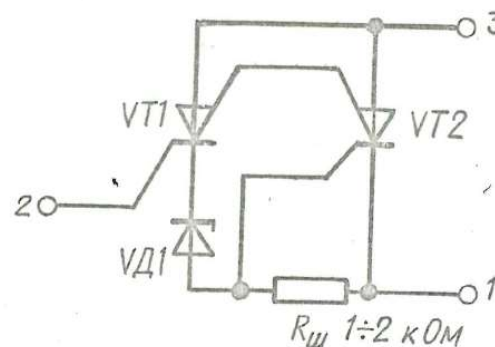
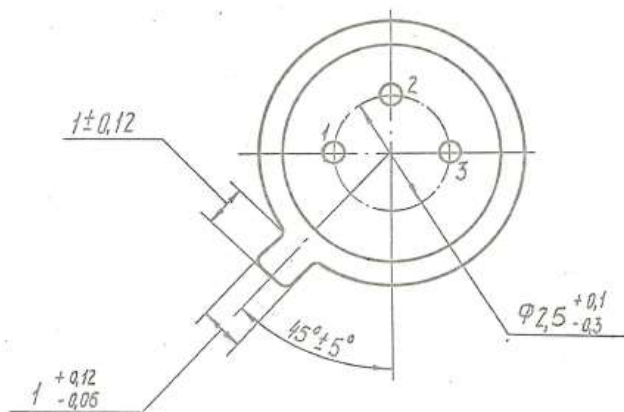


Integrated circuit series 520

Series integrated thyristor switch microcircuits 520, intended for use in special-purpose radio-electronic equipment.



Вид А



Номер вывода	Назначение вывода	Примечание
1	Катод	Нумерация выводов указана условно
2	Вход	
3	Анод	

Weight not more than 0,4 g



Integrated circuit series 520

Main electrical parameters

Type designation	Parameter name	Letter symbol	Min.	Max.	Supply A, V	Supply B, G	Max out current, mA	Temperature, °C
520KT1A-G	Output leakage current, μA	$I_{\text{leak, out}}$		10	15	30		25±10
				10	15	30		-60±2
				100	15	30		125±5
520KT1A-G	Input voltage, V	U_{in}	5	7	15	30	50	25±10
			5	7	15	30	50	-60±2
			5	7	15	30	10	125±5
520KT1A, B	Input current, μA	I_{in}		100	15	30	50	25±10
				120	15	30	50	-60±2
				100	15	30	10	125±5
520KT1V, G	Input current, μA	I_{in}		500	15	30	50	25±10
				550	15	30	50	-60±2
				500	15	30	10	125±5
520KT1A-G	Residual voltage, V	U_{res}		1,35			50	25±10
				1,5			50	-60±2
				1,35			10	125±5
520KT1A, V	Pulse output voltage, V	$U_{\text{out, p}}$	5		12	12		60±2
520KT1B, G	Pulse output voltage, V	$U_{\text{out, p}}$	46148		12	12		60±2
520KT1A-G	Max. permissible voltage rise rate in off state, V/us	dU/dt	10		15	15		125±5
520KT1A, V	Minimum output current, mA	$I_{\text{out, min}}$		5				60±2
520KT1B, G	Minimum output current, mA	$I_{\text{out, min}}$	0.5					125±5

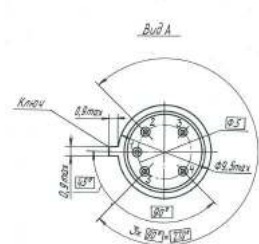
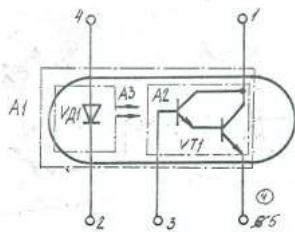
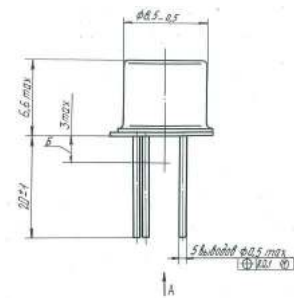
Weight not more than 0.4 g



Optocouplers



Transistor optocouplers 3OT131A



Поз. обозначение	Наименование
A1	Арматура ТТЗ.506.211
A2	Блок арматуры ТТЗ.516.163
VT1	Кристалл ТТЗ.344.403
A3	Блок арматуры ТТЗ.516.080-02
VD1	Кристалл ТТЗ.344.393-01@

Transistor optocouplers in a metal-glass package, consisting of silicon planar-epitaxial transistor receivers and mesa-epitaxial emitters based on GaAlAs compound, intended for switching DC circuits with galvanic isolation between input and output in special-purpose radio-electronic equipment

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Input voltage at ($I_{in}=10$ mA), V	U_{in}		1,7	25
	U_{in}		1,7	85
	U_{in}		2,0	-60
Output residual voltage, V ($I_{in}=2$ mA, $I_{out}=10$ mA) ($I_{in}=2$ mA, $I_{out}=7$ mA) ($I_{in}=2$ mA, $I_{out}=5$ mA)	$U_{out.res}$		1,5	25
	$U_{out.res}$		1,5	85
	$U_{out.res}$		1,5	-60
Output leakage current ($I_{in}=0$, $U_{sw}=15$ V), μ A	$I_{leak.out}$		10	25
	$I_{leak.out}$		100	85
	$I_{leak.out}$		10	-60
Insulation resistance, Ohm ($U_{ins}=500$ V)	R_{ins}	10^{11}		25

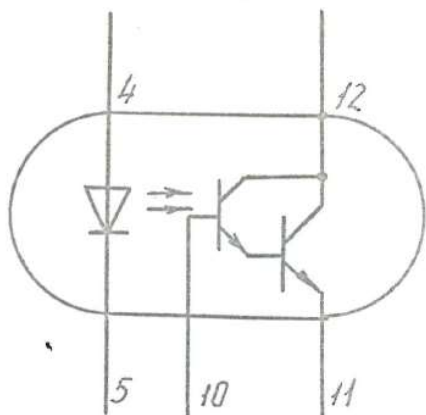
Weight not more than 2 g



Transistor optocouplers

3OT122A, 3OT122B, 3OT122V, 3OT122G

Электрическая схема оптопары



4,5,10,11,12 - выводы оптопары.

Transistor optocouplers, consisting of silicon epitaxial-planar transistor receiver and mesa-epitaxial infrared diode emitter, in a metal-glass package, intended for бесконтактной switching DC circuits with galvanic isolation between input and output in special-purpose radio-electronic equipment

Main electrical parameters

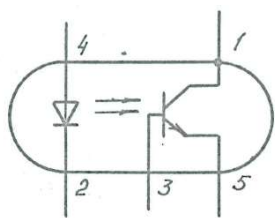
Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Input voltage at ($I_{in}=5$ mA), V	U_{in}	1,0	1,6	25 ± 10
	U_{in}	-	1,6	70 ± 3
	U_{in}	-	1,8	-60 ± 5
Output residual voltage, V $I_{in}=5$ mA, $I_{out}=15$ mA for 3OT122A, 3OT122V, 3OT122G $I_{in}=5$ mA, $I_{out}=7$ mA for 3OT122B	U_{res}		1,5	25 ± 10
	U_{res}		1,5	70 ± 3
	U_{res}		1,9	-60 ± 5
Output residual voltage, V $I_{in}=1$ mA, $I_{out}=1$ mA for 3OT122A $I_{out}=3$ mA for 3OT122V,G $I_{out}=5$ mA for 3OT122B	U_{res}		1,5	25 ± 10
	U_{res}			
	U_{res}			
Output leakage current, μA $I_{in}=0$ $U_{sw}=50$ V for 3OT122A $U_{sw}=30$ V for 3OT122B, 3OT122V $U_{sw}=15$ V for 3OT122G	$I_{leak.out}$		10	25 ± 10
	$I_{leak.out}$		100	70 ± 3
	$I_{leak.out}$		10	-60 ± 5
Insulation resistance, Ohm	R_{ins}	10^9		25 ± 10

Weight not more than 0,6 g



Transistor optocouplers

3OT126A, 3OT126B, AOT126A, AOT126B



вывод 1 – коллектор транзисторного приемника

вывод 5 – эмиттер транзисторного приемника

вывод 3 – база транзисторного приемника

вывод 2 – катод излучающего диода

вывод 4 – анод излучающего диода

вход оптопары – выводы 4, 2

выход оптопары – выводы 1, 5

Transistor optocouplers, consisting of silicon planar-epitaxial n-p-n transistor receivers and mesa-epitaxial emitters based on GaAlAs in a metal-glass package, intended for switching DC circuits with galvanic isolation between input and output in special-purpose radio-electronic equipment (in the case of 3OT126A,B) and national-economy products (in the case of AOT126A,B).

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Input voltage at ($I_{in}=20$ mA), V	U_{in}	2	2	25 ± 10
		2	2	85 ± 3
		2	2	-60 ± 3
Output residual voltage, V ($I_{in}=20$ mA, $I_{out}=10$ mA)	U_{res}	0,3	0,3	25 ± 10
		0,4	0,4	85 ± 3
		0,3	0,3	-60 ± 3
Output leakage current, μ A $U_{sw}=30$ V for 3OT126A, AOT126A $U_{sw}=15$ V for 3OT126B, AOT126B	$I_{leak.out}$	10	10	25 ± 10
		200	200	85 ± 3
		10	10	-60 ± 3
Insulation resistance, Ohm	R_{ins}	10^{11}	10^{11}	25 ± 10

Electrical parameters by group

Maximum permissible switched voltage, V

Symbol designation	3OT126A, AOT126A	3OT126B, AOT126B
$U_{KOM MAX}$	30	15

Weight not more than 2 g



Transistor optocouplers 3OT144A

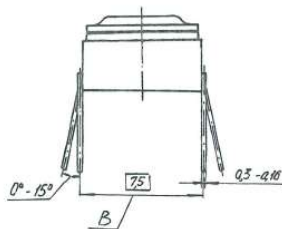
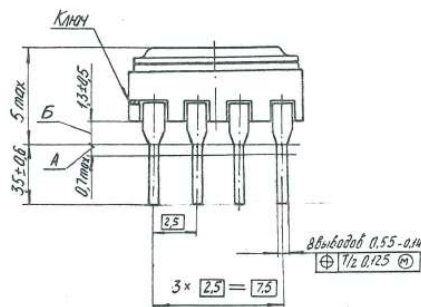
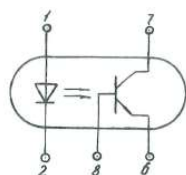


Схема электрическая принципиальная



Transistor optocoupler in a metal-ceramic package, consisting of silicon epitaxial-planar transistor receiver and epitaxial infrared diode emitter, intended for implementation of intra-unit and inter-unit high-voltage insulation in special-purpose radio-electronic equipment.

Main electrical parameters

Parameter name, unit of measurement, measurement mode	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Input voltage at ($I_{in}=10$ mA), V	U_{in}		1,8 2,0 1,8	25±10 -60±3 85±5
Output residual voltage, V $I_{in}=10$ mA, $I_{out}=10$ mA, $R_{be}=100$ kOhm $I_{in}=10$ mA, $I_{out}=5$ mA, $R_{be}=100$ kOhm $I_{in}=10$ mA, $I_{out}=7$ mA, $R_{be}=100$ kOhm	$U_{out.res}$		0,3 0,3 0,3	25±10 -60±3 85±5
Output leakage current at $U_{sw}=30$ V, $I_{in}=0$, $R_{be}=100$ kOhm, μ A	$I_{leak.out}$		10 100	25±10 -60±3 85±5
Insulation resistance at $U_{ins}=500$ V, Ohm	R_{ins}	10^{11}		25±10

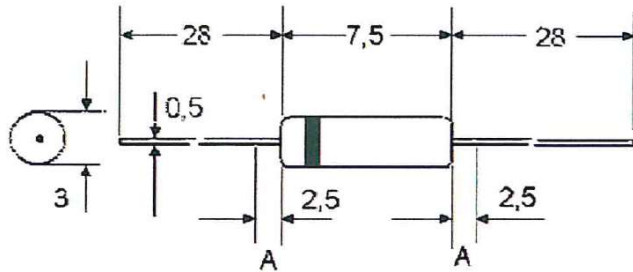
Weight not more than 2 g



Zener Diodes



Zener diodes 2S108A-Zh, I-N, P-S



Precision thermally compensated Zener diodes class 0,02 with nominal stabilization voltage 6,4 V, with specified low-frequency noise, in a glass package, intended for use as reference-voltage source in precision equipment.

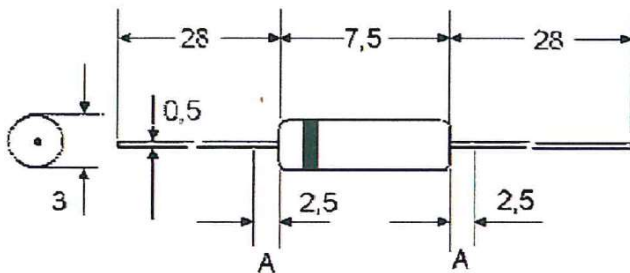
Main electrical parameters

Parameter name (measurement mode), unit of measurement	Symbol designation	min.	max.
Deviation of stabilization voltage from nominal value $U_{st} = 6,4$ V, %	ΔU_{CT}	-5	+5
Differential resistance, Ohm, at ambient temperature: +25°C 125°C	R_{st}		15 40
Differential resistance at minimum current 3 mA, Ohm	R_{st1}		70
Time instability of stabilization voltage over any 5000 hours within the total minimum operating life, mB at ambient temperature: from -5°C to 65°C from -60°C to +125°C	dU_{st1}	-1,3 -3,2	+1,3 +3,2
Temperature coefficient of stabilization voltage, averaged at temperature ambient environment from -5°C to +65°C, %/°C	αU_{CT}	-0,002	+0,002

Weight not more than 0,5 g



Zener diodes 2S108A-Zh, I-N, P-S



Precision thermally compensated Zener diodes class 0,02 with nominal stabilization voltage 6,4 V, with specified low-frequency noise, in a glass package, intended for use as reference-voltage source in precision equipment.

Electrical parameters by group

Temperature drift of stabilization voltage in the range ambient temperature from -5 °C to +65 °C, mV

Symbol designation	A, G, Zh	B, D, I, L, N	V, E, K, M, P, R, S
dUst θ	min.	min.	min.
	-9	-4,5	-2,5
	max.	max.	max.
	+9	+4,5	+2,5

Time instability of stabilization voltage over 1000 h within total minimum operating life, guaranteed by individual certificate, mV

Symbol designation	G, D, E	Zh, I, K	L, M	N, P	R	S
dUst2	min.	min.	min.	min.	min.	min.
	-0,64	-0,32	-0,13	-0,7	-0,035	-0,02
	max.	max.	max.	max.	max.	max.
	+0,64	+0,32	+0,13	+0,7	+0,035	+0,02

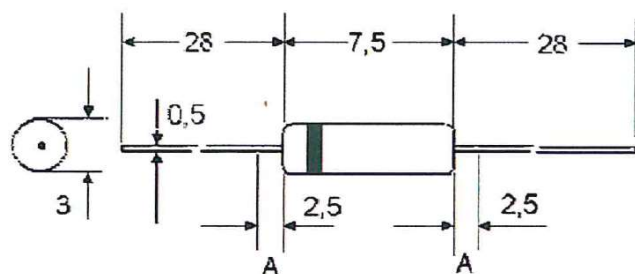
Low-frequency noise amplitude in frequency range from 0,01 to 1 Hz, μ V

Symbol designation	G, D, E, Zh, I, K	L, M, N, P,	R	S
Uinsm	max.	max.	max.	max.
	40	20	10	6

Weight not more than 0,5 g



Zener diodes 2S117



Precision thermally compensated Zener diodes class 0,02 types 2S117A,B,V and certified class 0,01 types 2S117G,D,E, class 0,005 types 2S117Zh,I,K, class 0,002 type 2S117L,M,N,P, in a glass package, intended for special-purpose equipment.

Main electrical parameters

Parameter name (measurement mode), unit of measurement	Symbol designation	min.	max.
Deviation of stabilization voltage from nominal value $U_{st} = 6,4 \text{ V}$, %	ΔU_{CT}	-5	+5
Differential resistance, Ohm, at ambient temperature: from -60°C to 60°C 125°C	R_{st}		20 30
Time instability of stabilization voltage over any 5000 hours within the total minimum operating life, mB at ambient temperature: from -5°C to 65°C from -60°C to +125°C	dU_{st1}	-1,3 -3,2	+1,3 +3,2

Electrical parameters by group

Temperature drift of stabilization voltage in the range ambient temperature from -5 °C to +65 °C, mV

Symbol designation	A, G, Zh,	B, D, I, L, R	V, E, K, M, N, P, S, T, U, F
$dU_{ST\Theta}$	min. -8 max. +8	min. -4 max. +4	min. -2 max. +2

Time instability of stabilization voltage over 1000 h within total minimum operating life, guaranteed by individual certificate, mV

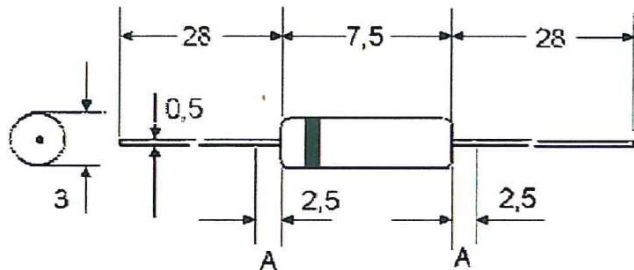
Symbol designation	G, D, E	Zh, I, K	L, M, N, P	R, S	T, U	F
dU_{st}	min. -0,7 max. +0,7	min. -0,35 max. +0,35	min. -0,14 max. +0,14	min. -0,7 max. +0,7	min. -0,035 max. +0,035	min. -0,02 max. +0,02

Weight not more than 0,5 g



Zener diodes KS108A, KS108B, KS108V

Precision Zener diodes in a metal-glass package, with nominal stabilization voltage 6,4 V and intended for use as reference-voltage source in precision equipment.



Maximum permissible electrical ratings in the range darkperatur environment:

Maximum permissible DC stabilization current at ambient temperature from -60°C to +60°C - 10 mA,

Maximum permissible DC stabilization current at ambient temperature at +125 °C - 7,5 mA,

Maximum permissible power dissipation at ambient temperature from - 60 °C to 60 °C - 70 mW,

Maximum permissible power dissipation at ambient temperature from 125 °C - 50 mW.

Main electrical parameters

Parameter name (measurement mode), unit of measurement	Symbol designation	KS108A		KS108B		KS108V	
		min.	max.	min.	max.	min.	max.
Deviation of stabilization voltage from nominal value $U_{st} = 6,4$ V, %	ΔU_{CT}	-5	+5	-5	+5	-5	+5
Differential resistance at $I_{st}=7,5\pm0,2$ mA, ambient temperature +25°C, at 125°C. Ohm	r_{st}		15 40		15 40		15 40
Temperature drift of stabilization voltage in the range ambient temperature from -5 °C to +60 °C, mV	$\Delta U_{CT\theta}$	-8,4	+8,4	-4,2	+4,2	-2,1	+2,1
Temperature coefficient of stabilization voltage, averaged at temperature ambient environment from -5°C to +60°C, %/°C	α_{UCT}	-0,002	+0,002	-0,001	+0,001	-0,0005	+0,0005
Nominal stabilization current, mA	I_{st}		7,5		7,5		7,5

Weight not more than 0,5 g

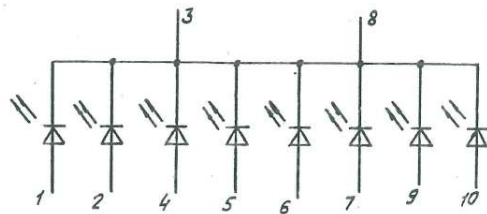


Digital Indicators

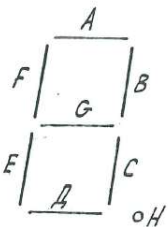


Digital indicators 3LS339B, KIPTs60A-1/7K

Gallium-aluminum arsenide mesa-epitaxial indicators digital red emission color 3LS339B and KIPTs60A-1/7K in glass-ceramic package, intended for visual indication in special-purpose equipment.



Номер вывода	Полярность
1	анод G
2	анод F
3	общий катод
4	анод E
5	анод D
6	анод C
7	анод H
8	общий катод
9	анод B
10	анод A



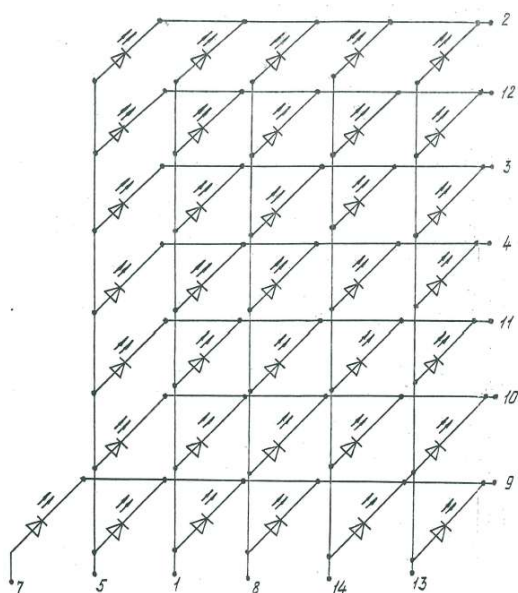
Name, unit of measurement	Designation	3LS339B		KIPTs60A-1/7K	
		Min.	Max.	Min.	Max.
Indicator luminous intensity (at forward current $I_{fwd}=3\text{mA}$ through the element), uccd	Iv	160		400	
DC forward voltage on each element, V	Ufwd		1,9		2,0

The indicators have seven segments and a decimal point, emitting light when forward current is applied. Different combinations of elements, provided by external switching, allow any digit from 0 to 9 to be displayed. Digit height 2,5 mm.



Digital indicators ZLS340B1, KIPTs11A-5kh7K

Gallium-aluminum arsenide mesa-epitaxial indicators digital red emission color ZLS340B1 and KIPTs11A-5kh7K in a plastic package, consisting of discrete elements, manufactured using epitaxial technology, intended for visual indication in special-purpose equipment (in case of ZLS340B1) and national economy applications, in including for export supply (in case of KIPTs11A-5kh7K)



Номер вывода	Ряд-колонка	Полярность
1	Колонка 2	+
2	Ряд 1	-
3	Ряд 3	-
4	Ряд 4	-
5	Колонка 1	+
6	отсутствует	-
7	точка	+
8	Колонка 3	+
9	Ряд 7	-
10	Ряд 6	-
11	Ряд 5	-
12	Ряд 2	-
13	Колонка 5	+
14	Колонка 4	+

Name, unit of measurement	Designation	ZLS340B1		KIPTs11A-5kh7K	
		Min.	Max.	Min.	Max.
Indicator luminous intensity (at forward current $I_{fwd}=10\text{mA}$ through the element), ucd	Iv	1,3		1,0	
DC forward voltage on each element (at forward current $I_{fwd}=10\text{mA}$ through the element), V	Ufwd		2,0		2,0

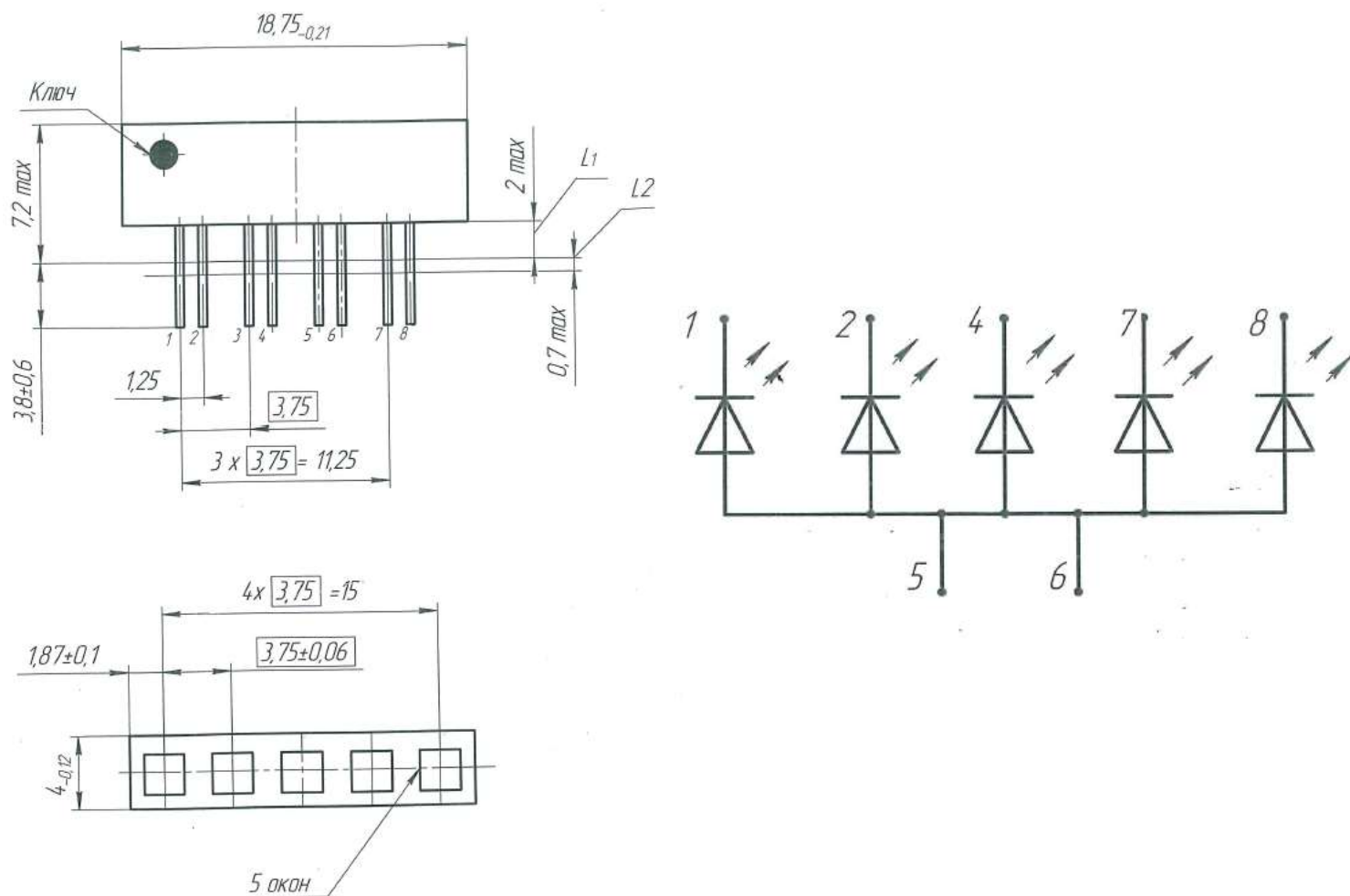
The indicators have 35 elements (7 rows po 5 elements in ryadu) and left decimal point, emitting light at application of forward current. Different combinations of elements, provided by external switching, allow any digit from 0 to 9 to be displayed and decimal point, as well as letters of the Russian and Latin alphabets and other possible characters.

Digit height ~ 9 mm



Bar-scale indicators KIPT21A-5K.

Character-generating indicators bar-scale semiconductor with red emission color of 5 discrete elements in a plastic package, intended for visual indication

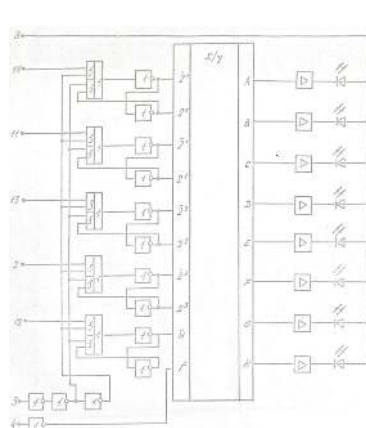


Parameter name, measurement mode, unit of measurement	Letter symbol	KIPT21A-5K		Temperature, °C
		Minimum	Maximum	
Constant forward voltage of display element (Ifwd=20mA), V	Vfwd		3,5 4,5 3,5	25 -25 70
Average luminous intensity of display element (Ifwd=20mA через элемент), ucd	I _{V avg}	400		25
Relative luminous-intensity spread between elements	$\frac{I V_{\max}}{I V_{\min}}$		3	25

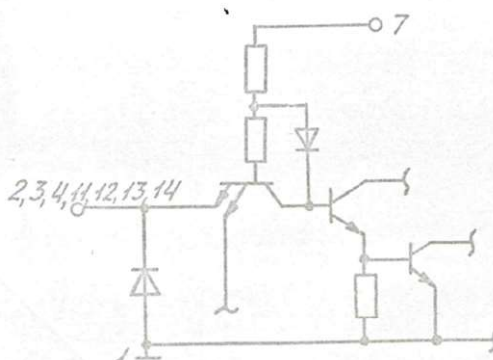


Digital indicators KM490IP2

Digitally controlled indicators red emission color, intended for use in radio-electronic equipment. The control circuit is made using CMOS technology. The indicators have 7 segments and decimal point, allow any digit to be displayed from 0 to 9 and decimal point



Электрическая схема выводов 2,3,4,11,12



Номер вывода	Назначение внешнего вывода
1	Общий
2	Вход 2 ³
3	Разрешение записи Р
4	Вход гашения Г
7	Питание
8	Питание индикации
11	Вход 2 ¹
12	Управление точкой
13	Вход 2 ²
14	Вход 2 ⁰

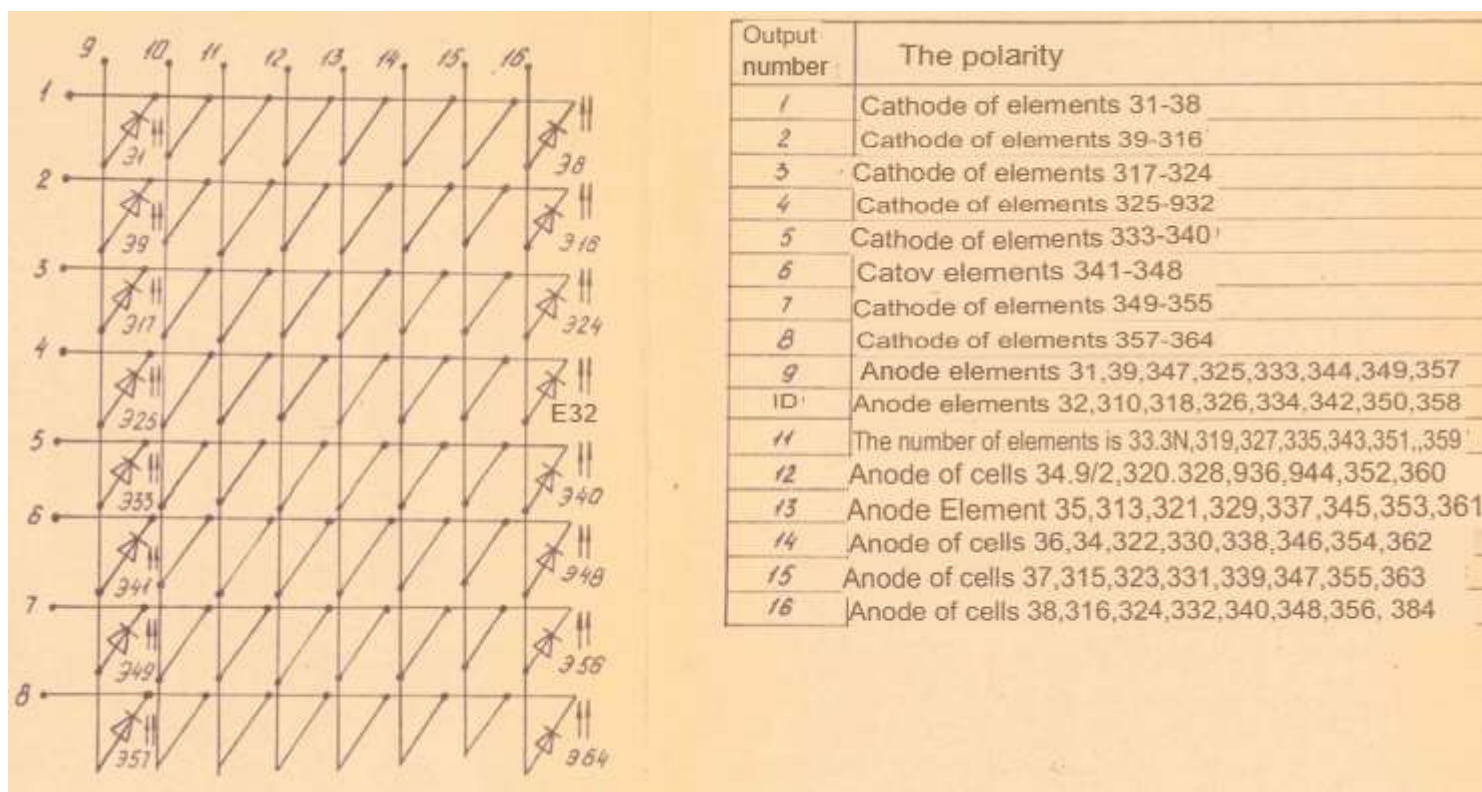
Main electrical parameters

Parameter name (measurement mode), unit of measurement	Symbol designation	Limit		Temperature, °C
		Min.	Max.	
Low-level input current at $U_{in}=0,4$ V, $U_{supply}=5,25$ V, μA	I^0_{vkh}	-	1,6 1,6 1,6	25±10 -45±3 70±3
High-level input current at $U_{in}=2,4$ V, $U_{supply}=5,25$ V, μA For all inputs except write-enable input (sinking)	I^1_{vkh}	-	100 250 250	25±10 -45±3 70±3
High-level input current at $U_{in}=2,4$ V, $U_{supply}=5,25$ V, μA For write-enable input (sinking)	I^1_{vkh}	-	200 250 250	25±10 -45±3 70±3
Current consumption at $U_{supply}=5,25$ V, μA	I_{cons}	-	75 80 80	25±10 -45±3 70±3
Display current at $U_{supply}=5,25$ V, $U_{ind}=3,0$ V, skhema in sostoyanii indication "8", μA	I_{disp}	-	160 190 190	25±10 -45±3 70±3
Luminous intensity at $U_{supply}=5,25$ V, $U_{ind}=3,0$ V, skhema in sostoyanii indication "8", ucd	I_p	4		25±10



Character-generating indicators IPG06A-8kh8K

Planar aluminum-gallium arsenide character-generating graphic semiconductor screen indicator modules-red emission color with number of elements 8kh8 size 10kh10 mm type IPG06A-8kh8K in a plastic package, intended for use in composite special-purpose information display systems

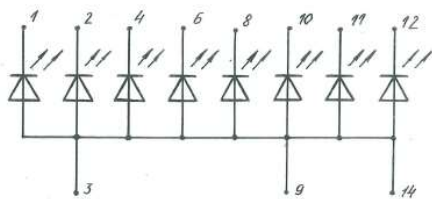


Main electrical parameters

Parameter name, measurement mode, unit of measurement	Symbol designation	IPG06A-8kh8K		Temperature, °C
		Min.	Max.	
DC forward voltage of display element, V	Ufwd		3,0 2,5 2,5	-60 25 70
Average luminous intensity of display element, uccd	Iv avg	0,6		25
Relative luminous intensity variation between elements	δIv+ δIv-		60 60	25

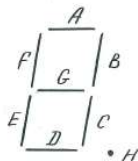


Single-digit digital indicators KLTs202V



Номер вывода	Полярность
1	Катод А
2	Катод F
3	Общий анод
4	Катод E
6	Катод H
8	Катод D
9	Общий анод
10	Катод C
11	Катод G
12	Катод B
14	Общий анод

Схема расположения
свещающихся элементов



Digital indicators red emission color KLTs202V (gallium aluminum arsenide) in a plastic package, consisting of discrete elements, manufactured using epitaxial-diffusion technology, intended for indication. The indicator has 7 segments and decimal point, emitting when forward current is applied. Different combinations of elements, provided by external switching, allow any digit from 0 to 9 to be displayed and decimal point

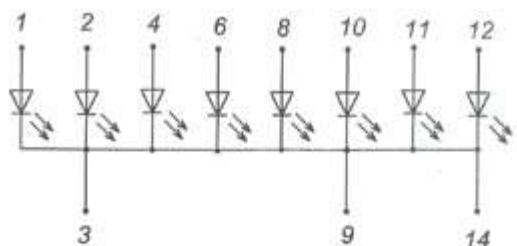
Main electrical parameters

Parameter name, measurement mode, unit of measurement	Letter symbol	KLTs202V		Temperature, °C
		Minimum	Maximum	
Average luminous intensity of display element (Ifwd=20mA через элемент), ucd	$I_{V \text{ avg}}$	0,5		
Average luminous intensity of decimal point (Ifwd=20mA), ucd	I_V	0,07		
Constant forward voltage of display element, V Ifwd=20mA Ifwd=20mA Ifwd=7,5mA	Vfwd		2,5 4 2,5	25±10 -45±3 70±3
Constant forward voltage of decimal point, V Ifwd=20mA Ifwd=20mA Ifwd=7,5mA	Vfwd		2,5 4,0 2,5	25±10 -45±3 70±3
Relative luminous-intensity spread between elements	$\frac{I_{V \text{ max}}}{I_{V \text{ min}}}$		3,0	25

Character height 20 mm
Indicator weight max. 6 g

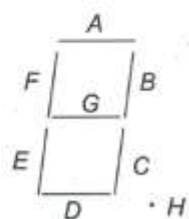


Single-digit digital indicators KLTs302G



Output number	The polarity
1	Anode A
2	Anode F
3	The cathode is common
4	Anode E
6	Anode H
8	Anode D
9	The cathode is common
10	Anode W
11	Anode G
12	The anode is I
14	The cathode is common

The layout of the luminous elements



Character indicators gallium phosphide green emission color, consisting of discrete elements, manufactured using epitaxial-diffusion technology, intended for visual indication in general-purpose equipment. Digit height 20 mm. The indicators have 7 segments and decimal point. Different combinations of elements allow any digit from 0 to 9 to be displayed and decimal point

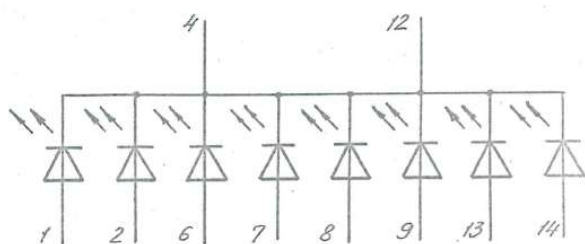
Main electrical parameters

Name, unit of measurement	Designation	Limit		Temperature, °C
		Minimum	Maximum	
Average luminous intensity of display element, ucd	$I_v(av)$	400		25
Luminous intensity of decimal point, ucd	I_v	100		25
Constant forward voltage of element, V	UF		3,5 5,0 3,5	25 -25 70
Relative luminous-intensity spread between elements	$\frac{I V_{\max}}{I V_{\min}}$		3	25

The indicators are made in a plastic package on a fiberglass laminate board. Indicator weight type KLTs302G max. 6 g

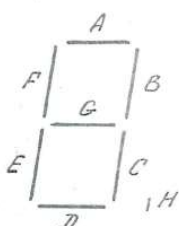


Single-digit digital indicators ZLS338A2, ZLS338B2



Номер вывода	Полярность
1	Анод F
2	Анод G
4	Катод общий
6	Анод E
7	Анод D
8	Анод C
9	Анод H
12	Катод общий
13	Анод B
14	Анод A

Схема расположения светящихся элементов



Gallium phosphide-arsenide digital indicators green emission color, in a plastic package type KI5-4, consisting of discrete elements, manufactured using epitaxial-diffusion technology, intended for visual indication.

ZLSZZ8A2 - with common cathode; ZLSZZ8B2 - with common anode.

Main electrical parameters

Parameter name, measurement mode, unit of measurement	Letter symbol	ZLS338A2, ZLS338B2	
		Minimum	Maximum
Average luminous intensity (Ifwd=20mA cherez element), mcd segment	$I_{V \text{ avg}}$	0,45	
Luminous intensity of decimal point (Ifwd=20mA cherez element), mcd	$I_{V \text{ avg}}$	0,15	
Constant forward voltage of each element (Ifwd=20mA), V	Ufwd		3,5
Relative luminous-intensity spread between indicator segments	$\frac{I_{V \text{ max}}}{I_{V \text{ min}}}$		3

Character height 7 mm.
Indicator weight max. 2,5 g.



New Developments



New developments Avalanche Photodiodes



Indium-gallium-arsenide-based avalanche photodiode

IFDL01A/ IFDL01A1

Application

- Range finding
- Optical data transmission
- Detection of weak optical signals

Structure / Characteristics

Crystal material	APD package	Photosensitive-area diameter, mm
InGaAs	TO-5/TO-46	0,2



Typical parameters

Spectral range λ , μm	0,9-1,7
Peak wavelength of relative spectral sensitivity characteristic λ , μm	1,55
Breakdown voltage U_{br} , V	40 ($I_{dark}=10\text{ }\mu\text{A}$)
Multiplication factor M, rel. units	20 ($I_{dark}=1\text{ }\mu\text{A}$)
Dark current I_d , nA	25 ($M=10$)
Ampere-watt sensitivity S, A/W	0,9 ($\lambda=1,55\text{ }\mu\text{m}$)
Bandwidth Δf , GHz	0,7 ($M=10$)
Capacitance C, pF	3 ($0,9 \cdot U_{br}, f=1\text{ MHz}$)

APD connection diagram
1 - Anode (+)
2 - Cathode (-)
3 - Not connected
Key mark shown

Current-voltage characteristic
Light current / Dark current
 I , A vs U , V

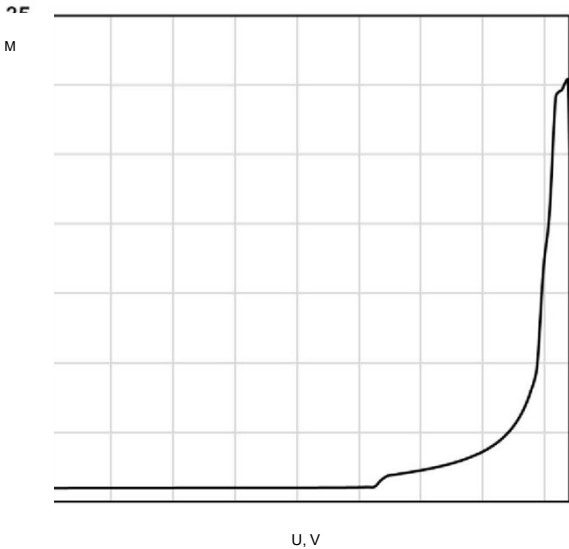
Current-voltage characteristic at optical radiation power 1 μW , $I(U_{rev})$



Indium-gallium-arsenide-based avalanche photodiode

IFDL01A/ IFDL01A1

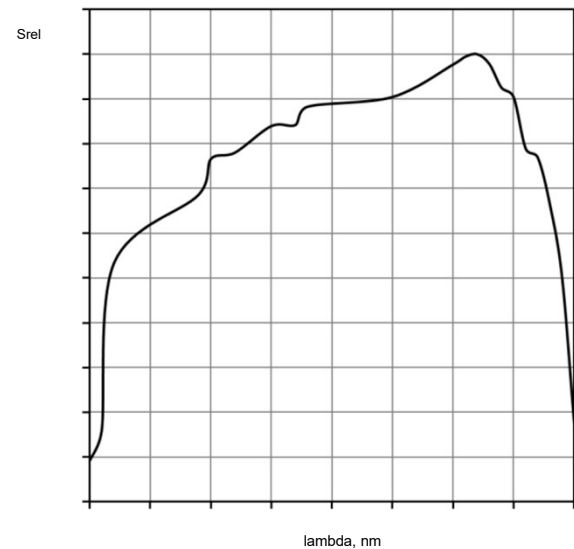
Main electrical parameters



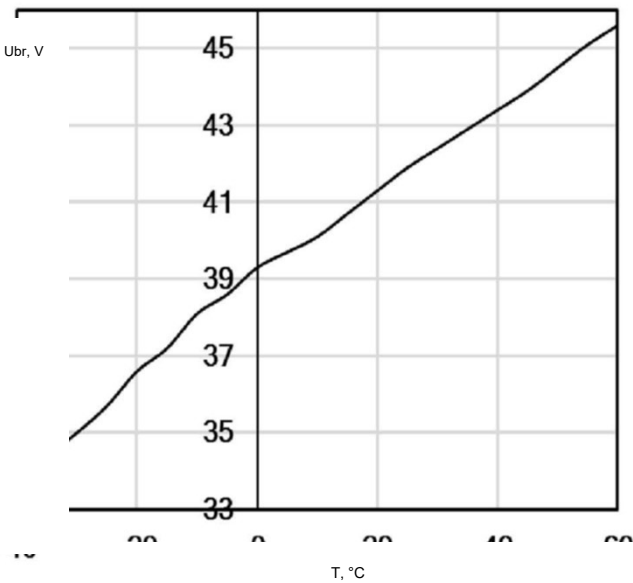
Multiplication factor at optical radiation power 1 uW, $M(U_{rev})$



Capacitance-voltage characteristic, $C(U)$



Relative spectral sensitivity characteristic, $S_{rel}(\lambda)$



Breakdown voltage dependence on temperature, $U(T)$



Silicon-based avalanche photodiode KFDL01A

Application

- Range finding
- Optical data transmission
- Detection of weak optical signals



Structure / Characteristics

Crystal material	APD package	Photosensitive-area diameter mm
Si	TO-5	0,5

APD connection diagram

- 1 - Anode (+)
 - 2 - Cathode (-)
 - 3 - Not connected
- Key mark shown

Typical parameters

Spectral range λ , μm	0,4-1,1
Peak wavelength of relative spectral sensitivity characteristic λ , μm	0,9
Breakdown voltage U_{br} , V	160 ($I_{dark}=10 \text{ uA}$)
Multiplication factor M, rel. units	70
Dark current I_{dark} , nA	1 ($M=10$)
Bandwidth Δf , GHz	0,6
Capacitance C, pF	2 ($0,9 \cdot U_{br}, f=1 \text{ MHz}$)

Current-voltage characteristic

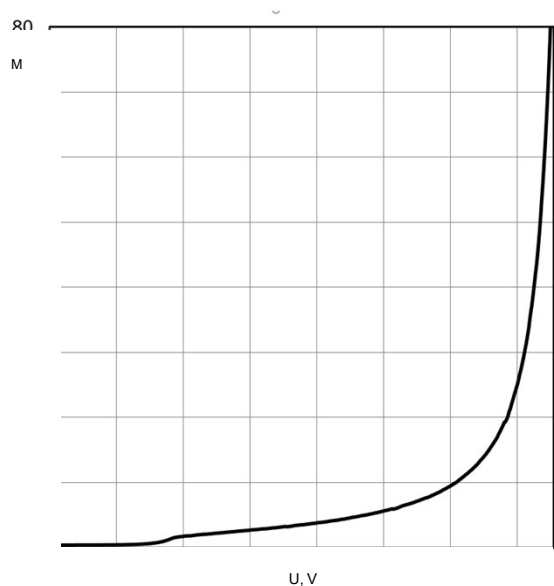
Light current / Dark current
 I , A vs U , V

Current-voltage characteristic at optical radiation power 1 μW , $I(U_{rev})$

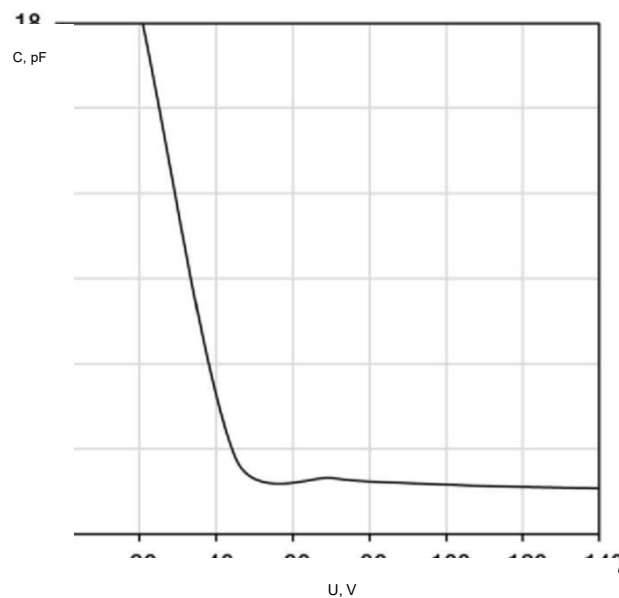


Silicon-based avalanche photodiode KFDL01A

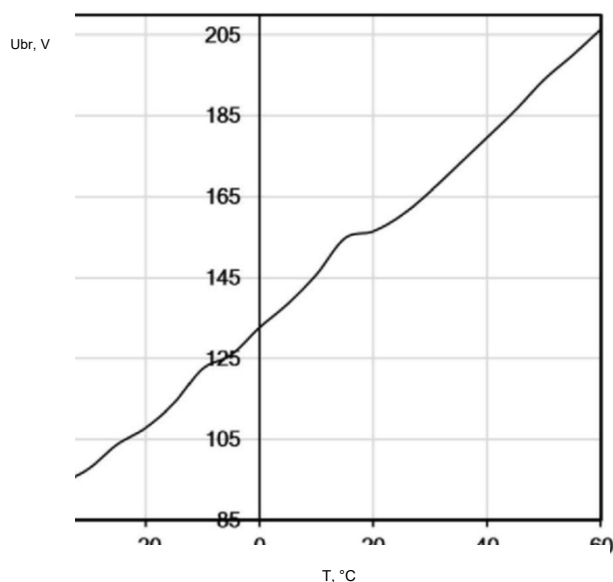
Main electrical parameters



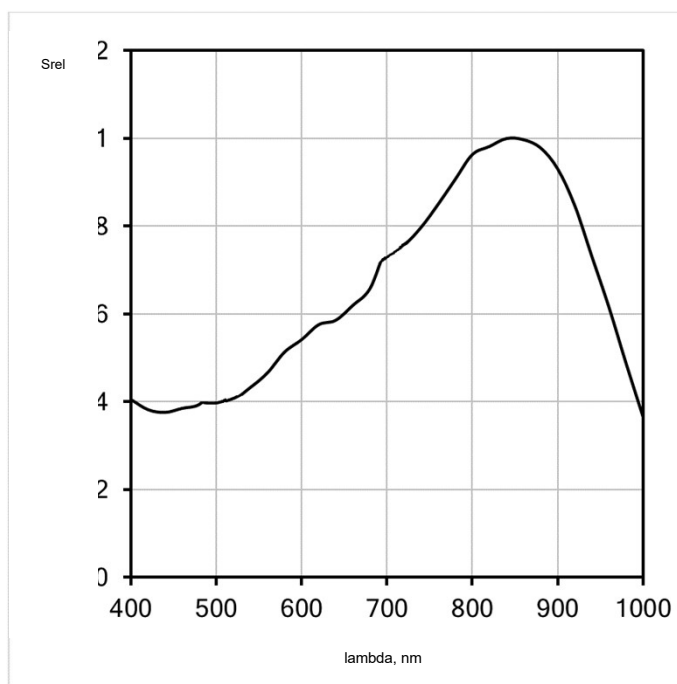
Multiplication factor at optical radiation power 1 μ W, $M(U_{rev})$



Capacitance-voltage characteristic, $C(U)$



Breakdown voltage dependence on temperature, $U(T)$



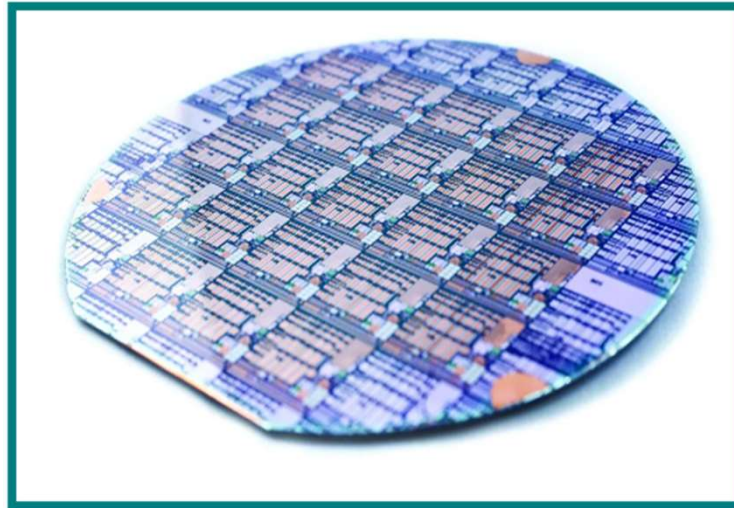
Relative spectral sensitivity characteristic, $S_{rel}(\lambda)$



New developments Heterostructures for LEDs and laser diodes



Heterostructures for LEDs and laser diodes



Technical Specifications: AlInGaP/GaAs and InGaN/Al2O3

Wavelength λ (λ_{dom} , nm)	Material	Optron JSC (lv pri 20mA)	Foreign analogues (Taiwan/China)	Voltage Drop (U_{np})
638 $\pm$ 5 nm	AlInGaP/GaAs	100	100	<2.2 B
625 $\pm$ 5 nm	AlInGaP/GaAs	140	140÷160	\$2.2 B
590 $\pm$ 5 nm	AlInGaP/GaAs	100	100÷110	<2.2 B
525 $\pm$ 10 nm	InGaN/Al2O3	30	30	3.0+3.2 B
460 $\pm$ 10 nm	InGaN/Al2O3	40	40	<3.2 B

Full compliance with Optotech, Epistar, and EpiTop parameters.

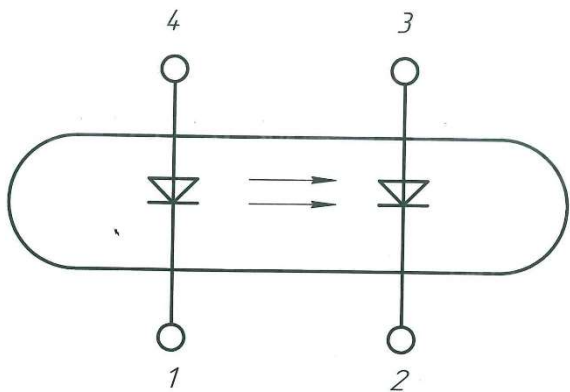
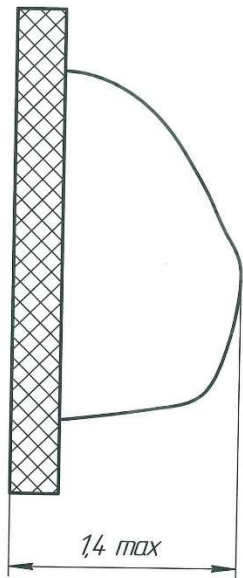
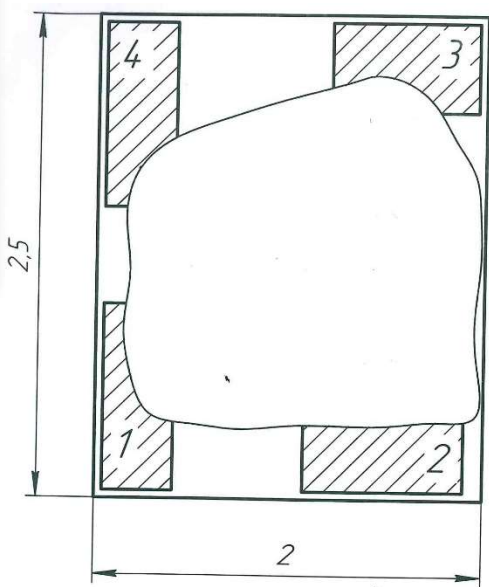
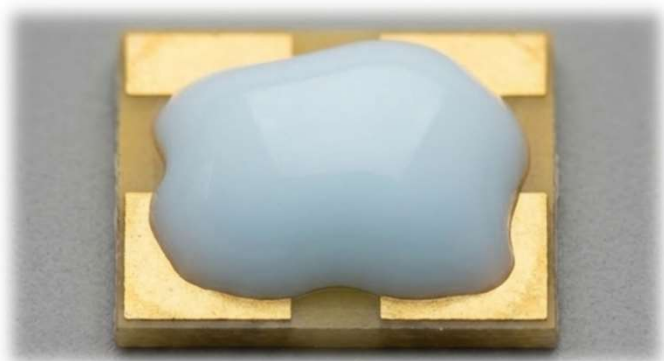


New developments

Diode optocoupler



Diode optocoupler AOD187A-6



Main parameters

Parameter name, unit of measurement	Symbol designation	Parameter value
Output-signal rise/fall time at Urev=10 V, lin. and=10 mA, ns, max.	t_{rise}/t_{fall}	100/100
Current transfer ratio at Urev=10 V, lin=10 mA, %, min.	K_I	1
Insulation voltage, V, min.	Uins	1500
Input voltage (at current 10 mA), V, max.	Uin	1,5
Reverse output voltage, V	$U_{out. rev}$	30
Feedthrough capacitance (Uc=0), pF max.	S_{fwd}	0,5





New developments

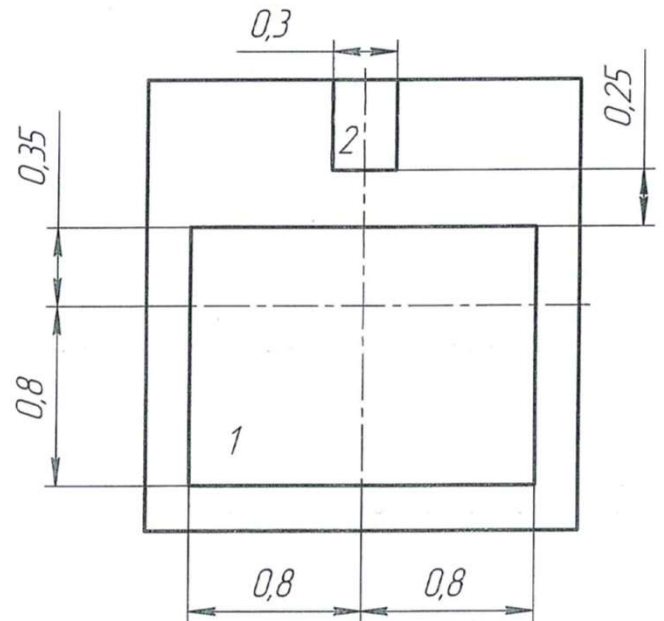
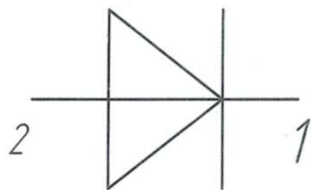
Microwave diodes in DFN

(Dual Flat No-leads)

package



Microwave diodes in DFN (Dual Flat No-leads) package



Miniaturization: Significant reduction in the dimensions and weight of end products.

Improved microwave properties: Minimization of parasitic inductances and capacitances due to the absence of long external leads.

Efficient heat removal: Excellent thermal characteristics thanks to the central exposed thermal pad.

Import independence: Creation of a reliable domestic component base of global level.

Target markets: Radar equipment, 5G/6G communication systems, aerospace industry, and high-precision measuring equipment.



New developments

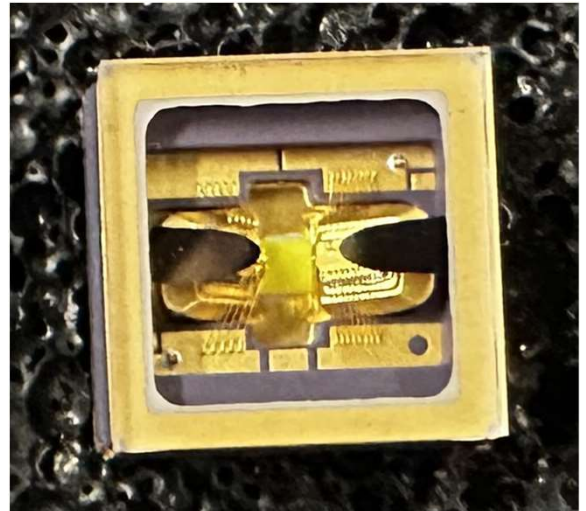
Laser diode in a package for SMD mounting



Laser diode in a package for SMD mounting

Application

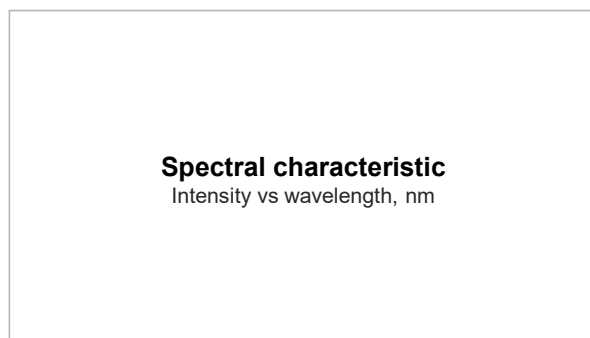
- Laser grow lamps
- Architectural lighting
- Special lighting
- Industrial lighting
- Automotive, aviation, railway and marine lighting



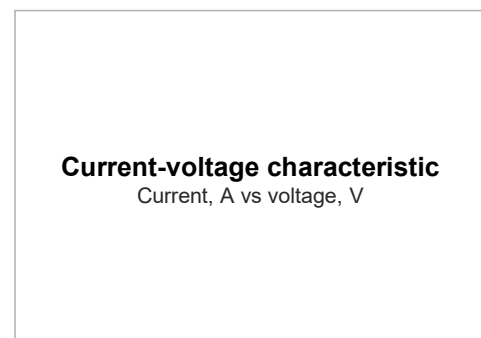
Parameters of white-light emitter in SMD package

No.	Name indicator/characteristic	Symbol	Value	
			min.	max.
1	Emission wavelength	nm	430	470
2	Radiation power in continuous mode	W	0,1	0,5
3	Operating current	A	-	1
4	Voltage at operating current	V	3,2	5,5
5	Emission color		White	
6	Luminous flux at operating current	lm	400	450
7	Brightness	cd/mm ²	1000	-
8	Color temperature (CCT)	K	5000	7000
9	Color rendering index	CRI	70	-

Spectral characteristic

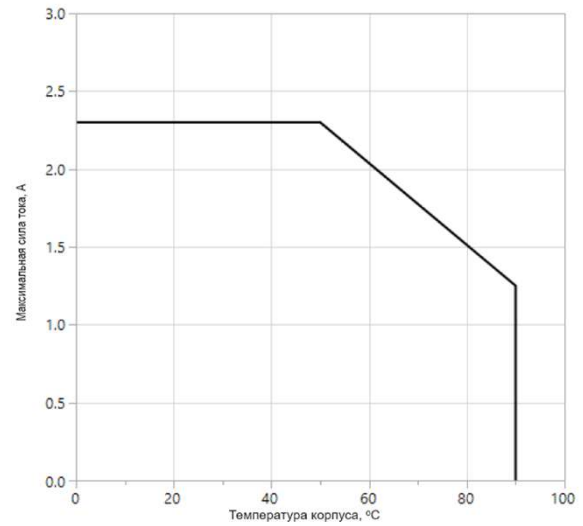
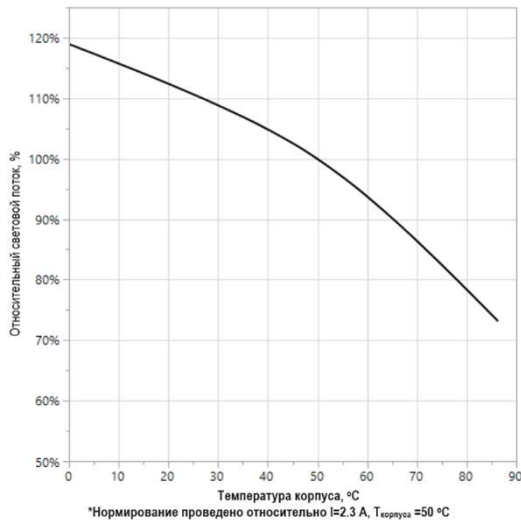


Current-voltage characteristic

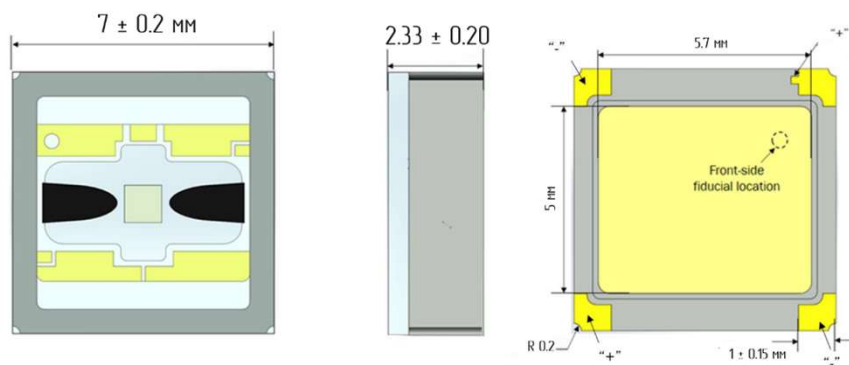


Laser diode in a package for SMD mounting

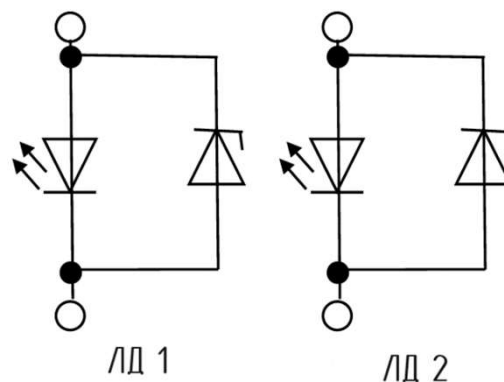
Temperature characteristics



Overall drawing



Electrical schematic



New developments Receiving optical microassembly



Receiving optical microassembly with PIN photodiode

Application:

- Telecommunication and 10 Gbit/s reception systems
- Optical data transmission



Product	Parameter name, unit of measurement	Designation	Value	Tolerance
10G PIN ROSA	Relative spectral sensitivity characteristic, nm	S(λ)	From 1100 to 1600	± 50
	Frequency bandwidth, GHz	B	7	± 2
	Sensitivity, dBm	R	-15	± 5
	Maximum optical power, dBm	Rmax	-1	± 1
	Transimpedance-amplifier supply voltage, V	Vcc	Max. 4	-
	Transimpedance-amplifier supply current, mA	Icc	Max. 50	-
	Operating temperature range	Ts	-40 °C ... +80 °C	± 5

Microassembly of a p-i-n photodiode and a transimpedance amplifier with automatic gain control and average optical power monitoring function in accordance with the SFF-8431 specification. The microassembly is based on a TO46 package with fiber output, intended for use with an LC/UPC connector and a single-mode optical fiber with a 3 mm jacket diameter, 125 μ m fiber and 9 μ m core diameter.

Gain response

Gain, dB vs frequency, GHz

Spectral sensitivity

Srel vs λ , nm

Gain dependence on operating frequency

Relative spectral sensitivity characteristic, Srel(λ)



Receiving optical microassembly with avalanche photodiode

Application:

- Telecommunication and 10 Gbit/s reception systems
- Optical data transmission
- Detection of weak optical signals

Features:

- High-sensitivity avalanche photodiode



Product	Parameter name, unit of measurement	Designation	Value	Tolerance
10G APD ROSA	Relative spectral sensitivity characteristic, nm	S(λ)	From 1100 to 1600	± 50
	Frequency bandwidth, GHz	B	7	± 2
	Sensitivity, dBm	R	-19	± 5
	Maximum optical power, dBm	Rmax	-5	± 1
	Transimpedance-amplifier supply voltage, V	Vcc	Max. 4	-
	Avalanche photodiode supply voltage, V	Vapd	Max. 55	-
	Transimpedance-amplifier supply current, mA	Icc	Max. 50	-
	Operating temperature range	Ts	-40 °C ... +80 °C	± 5
Microassembly of an avalanche photodiode and a transimpedance amplifier with automatic amplifier control and average optical power monitoring function in accordance with the SFF-8431 specification. The microassembly is based on a TO46 package with fiber output, intended for use with an LC/UPC connector and a single-mode optical fiber with a 3 mm jacket diameter, 125 μ m fiber and 9 μ m core diameter.				

Gain response
Gain, dB vs frequency, GHz

Gain dependence on operating frequency

Spectral sensitivity
Srel vs λ , nm

Relative spectral sensitivity characteristic, Srel(λ)



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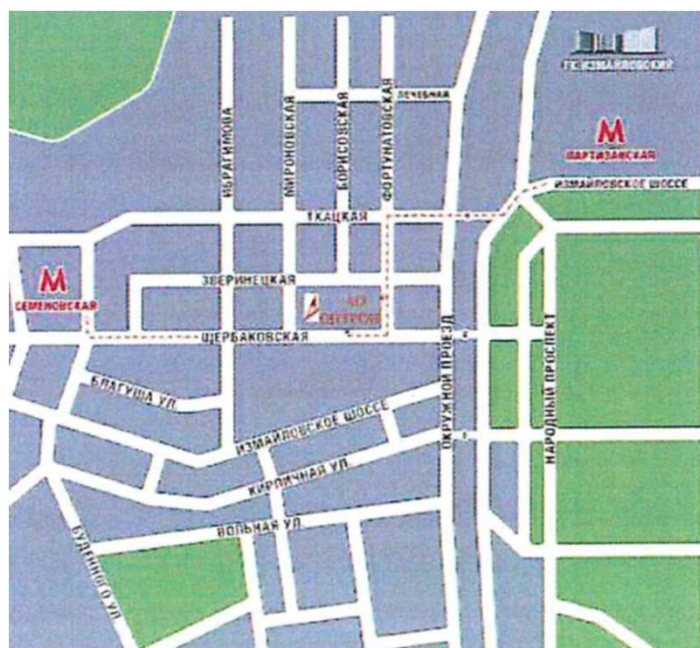
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Entrance to the enterprise territory is
from Shcherbakovskaya Street.

